

A CASE-STUDY BASED REPORT

ON

Crop Yield Prediction Using Random Forest, XGBoost, and Support Vector Regression

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by

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ABSTRACT:

Agriculture is one of the most important sectors as it provides food and supports the economy. Farmers often face difficulties in estimating crop yield because it depends on various factors such as rainfall, temperature, humidity, soil type, fertilizer usage, irrigation, and crop type. Incorrect estimation may result in poor planning and financial loss.

This project aims to predict crop yield using machine learning algorithms such as Random Forest, XGBoost, and Support Vector Regression (SVR). The project uses a Crop Yield Prediction dataset that contains agricultural and environmental information, including temperature, rainfall, humidity, soil type, crop type, fertilizer usage, irrigation, cultivated area, and crop yield. The collected data is cleaned and preprocessed before training the models to improve prediction accuracy. The performance of the models is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. The model with the best performance is selected for crop yield prediction. The proposed system helps farmers make better decisions related to crop selection, irrigation, fertilizer usage, and resource management. It can also support agricultural planning by providing an estimate of crop production before harvesting.

INTRODUCTION:

Agriculture plays a major role in the development of every country by providing food and employment. However, crop production is influenced by many factors such as weather conditions, rainfall, temperature, soil quality, irrigation, and fertilizer usage. Since these factors change frequently, farmers often find it difficult to estimate the expected crop yield before harvesting. This motivated us to work on a machine learning-based project that can predict crop yield more accurately using historical agricultural data.

In this project, we use Random Forest, XGBoost, and Support Vector Regression (SVR) to build crop yield prediction models. The dataset contains different agricultural and environmental factors that affect crop production. After preprocessing the dataset, the models are trained and tested, and their performance is compared using MAE, MSE, RMSE, and R^2 Score. The best-performing model will be used to predict crop yield. This project can help farmers and agricultural organizations make better decisions related to farming and resource management.

BACKGROUND:

Agriculture plays a vital role in food production and the economy, while crop yield is influenced by factors such as rainfall, crop type, cultivated area, production, fertilizer, and pesticide usage. Traditional yield estimation can be time-consuming, so this project applies Machine Learning techniques to predict crop yield using historical agricultural data. Random Forest, XGBoost, and Support Vector Regression (SVR) are used and compared using MAE, RMSE, and R^2 Score to evaluate their prediction performance.

The system provides a data-driven approach for estimating crop yield efficiently and consistently. It can also help support agricultural planning, resource management, and informed decision-making.

MOTIVATION OF THE PROJECT:

The motivation for this project is to develop an **efficient and data-driven approach for crop yield prediction**. Traditional methods can be time-consuming and may not effectively handle the large amount of agricultural data available today. By using **Machine Learning models such as Random Forest, XGBoost, and SVR**, the project aims to identify patterns in agricultural factors and provide reliable yield predictions. The system can support **better crop planning, resource management, and informed agricultural decision-making**. It can also help **reduce uncertainty in agricultural planning and enable more informed decisions before the harvesting stage**.

OVERVIEW OF THE WORK CARRIED OUT:

The project involved collecting and preprocessing a crop yield dataset containing agricultural and environmental factors such as crop, state, season, area, production, rainfall, fertilizer, and pesticide usage. Exploratory Data Analysis (EDA) was performed to understand the dataset and identify patterns. Three machine learning models—Random Forest, XGBoost, and Support Vector Regression (SVR)—were trained for crop yield prediction and evaluated using MAE, MSE, RMSE, and R^2 Score. A multi-layer web application was developed using React.js, Node.js, Python FastAPI, and Firebase Firestore to provide predictions and store prediction history.

SCOPE AND RESPONSIBLE-USE STATEMENT:

The scope of this project is to develop a machine learning-based crop yield prediction system using historical agricultural data and models such as Random Forest, XGBoost, and SVR. The system can support crop planning, resource management, and agricultural decision-making by providing data-driven yield estimates. However, predictions are dependent on the quality, accuracy, and relevance of the available data and should not be treated as guaranteed outcomes. The results are intended to assist decision-making rather than replace farmers' knowledge, expert advice, or real-world agricultural assessment.

PROBLEM STATEMENT:

Traditional crop yield estimation methods may not provide accurate and timely predictions. Crop yield depends on multiple factors such as rainfall, fertilizer, pesticide, cultivated area, crop type, season, and production, making prediction complex. Farmers and agricultural organizations may face difficulties in estimating productivity before harvesting, while analyzing large agricultural datasets manually is time-consuming. Therefore, there is a need for a machine learning-based system that can efficiently predict crop yield and compare different models to evaluate their prediction performance.

CASE STUDY SCENARIO:

An agricultural planning team wants to estimate the expected yield of crops before the harvesting season so that farmers can plan resources more effectively. For example, a farmer cultivating **rice in a particular Indian state** needs to consider factors such as the cultivated area, expected rainfall, fertilizer and pesticide usage, crop season, and previous production patterns. Instead of relying only on traditional estimation methods, the team uses historical agricultural data to build machine learning models that can identify patterns between these factors and crop yield. Similar research has demonstrated the use of agricultural, rainfall, and environmental data for district- and state-level crop-yield prediction in India.

CENTRAL CASE STUDY QUESTION:

A farmer planning to cultivate rice during the Kharif season needs to estimate the expected crop yield before harvesting. The expected yield can be affected by factors such as cultivated area, previous production, annual rainfall, fertilizer usage, pesticide usage, crop type, state, and season. Relying only on traditional estimation and past experience may make it difficult to analyze all these factors together. Therefore, a machine learning-based approach is proposed to analyze historical agricultural data and predict crop yield, helping farmers and agricultural planners make better-informed decisions about resource allocation and crop management.

1. What factors have the greatest influence on crop yield?
2. How can historical agricultural data be used to predict future crop yield?
3. Which machine learning model—Random Forest, XGBoost, or SVR—provides better prediction performance?
4. How accurately can the models predict crop yield?
5. How can crop-yield predictions support resource planning and agricultural decision-making?
6. What are the limitations of using historical data for crop-yield prediction?

FORMAL PROBLEM DEFINITION:

The objective of this study is to develop a machine learning-based crop yield prediction system that estimates crop yield using historical agricultural data. Given input features such as Year, State, Crop, Season, Area, Production, Annual Rainfall, Fertilizer, and Pesticide, the system predicts the corresponding Yield as a continuous numerical value. The study implements and compares Random Forest, XGBoost, and Support Vector Regression (SVR) models using regression metrics such as MAE, MSE, RMSE, and R^2 Score. The final system aims to provide a data-driven prediction that can support agricultural planning, resource management, and decision-making.

OBJECTIVES THAT GUIDE THE STUDY:

The main objectives of the study are to analyze historical crop-yield data and identify important agricultural factors that influence crop yield. The study aims to preprocess and explore the dataset, develop crop-yield prediction models using Random Forest, XGBoost, and Support Vector Regression (SVR), and evaluate their performance using MAE, MSE, RMSE, and R^2 Score. It also aims to develop a practical prediction system that can provide data-driven support for agricultural planning, resource management, and decision-making.

ASSUMPTIONS AND LIMITATIONS:

The study assumes that the historical crop-yield dataset is representative of agricultural conditions and that the recorded values are suitable for model training and prediction. The dataset contains 17,400 records, 10 columns, and no missing values, but the prediction is limited to the features available in the dataset. Factors such as soil quality, irrigation methods, temperature, humidity, and unexpected weather conditions are not included. Therefore, model predictions may differ from actual crop yield and should be considered decision-support estimates rather than guaranteed results.

DATASET AND EXPLORATORY DATA ANALYSIS

ABOUT THE DATASET:

The study uses a Crop Yield Prediction Dataset containing 17,400 records and 10 columns. It consists of 9 input features—Year, State, Crop, Season, Area, Production, Annual Rainfall, Fertilizer, and Pesticide—and one target variable, Yield. The dataset provides agricultural and environmental information used to analyze crop-yield patterns and train machine learning models for yield prediction. The dataset contains 0 missing values and 0 duplicate records.

PURPOSE OF EDA:

The purpose of Exploratory Data Analysis (EDA) is to understand the structure and quality of the crop-yield dataset before applying machine learning models. It involves examining the dataset dimensions, feature types, missing values, duplicate records, categorical features, numerical feature distributions, and Yield distribution. EDA also helps identify relationships between agricultural features and crop yield, supporting better feature preparation and model development.

PREPARATION STEPS ACTUALLY PERFORMED:

The dataset was first inspected to understand its dimensions, columns, and data types. Data quality checks were then performed to identify missing values and duplicate records. The categorical features State, Crop, and Season were analyzed, while numerical features such as Area, Production, Annual Rainfall, Fertilizer, and Pesticide were examined. Finally, the Yield target variable was analyzed through its distribution and relationships with other numerical features to prepare the dataset for machine learning.

EDA EXPLORATION – IMPLEMENTATION:

The EDA was implemented by examining the distribution of numerical agricultural features, including Production and Annual Rainfall, using graphs and boxplots. Correlation analysis was performed to study the relationships between numerical features and the Yield variable. Categorical features such as State, Crop, and Season were also analyzed to understand the structure of the agricultural data. These visualizations helped identify feature patterns, variations, and relationships that support the preparation of the dataset for machine learning.

SUMMARY OF EDA FINDINGS:

The EDA showed that the dataset contains **17,400 crop-yield records with no missing values or duplicate records**, making it suitable for further analysis and model training. The analysis identified variation in important agricultural features such as **Production, Annual Rainfall, Area, Fertilizer, and Pesticide**. Correlation analysis was used to examine the relationships between numerical features and **Yield**, while distribution plots and boxplots helped understand feature variation. Overall, the EDA provided a clear understanding of the dataset and supported the preparation of features for **Random Forest, XGBoost, and SVR** models.

ALGORITHM SELECTION AND IMPLEMENTATION

Three machine learning regression algorithms—**Random Forest, XGBoost, and Support Vector Regression (SVR)**—were selected for crop yield prediction. The preprocessed agricultural data was used to train these models, with **Yield** as the target variable. Random Forest was used to capture complex relationships between agricultural factors, XGBoost was used as a boosting-based regression approach, and SVR was used to model non-linear relationships. The models were evaluated and compared using **MAE, MSE, RMSE, and R² Score** to assess their prediction performance.

CONCLUSION

This study developed a machine learning-based crop yield prediction system using historical agricultural data. The dataset was explored and prepared through EDA, followed by the implementation of Random Forest, XGBoost, and SVR models. Their performance was evaluated using MAE, MSE, RMSE, and R² Score to compare prediction accuracy. The study demonstrates that machine learning can provide a data-driven approach to crop yield estimation, which can support agricultural planning, resource management, and informed decision-making.